
Gauss–Legendre Quadrature is a Drop-in Equivariance Upgrade for EquiformerV2 with Free Inference Speedup at Matched Accuracy

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Abstract

EquiformerV2 and other recent $SO(3)$ -equivariant networks for atomistic modeling apply pointwise nonlinearities by mapping spherical-harmonic (SH) coefficients to a discrete grid on the sphere, applying a scalar activation, and integrating back via quadrature. The standard implementation, inherited from `e3nn`, uses an equiangular Driscoll–Healy (DH) grid. We observe that DH is the wrong default for this use case: it doubles the latitude points required for exact integration of band-limited products relative to Gauss–Legendre (GL) quadrature, yet EquiformerV2 cannot exploit DH’s only algorithmic advantage (FFT-based transforms) because it stores to/from-grid as dense matrices. The result is an avoidable equivariance penalty. We implement a drop-in `CustomS03Grid` that replaces DH with GL inside any pretrained EquiformerV2 checkpoint. On a 12-layer / 128-channel / $\ell_{\max}=6$ model on QM9, the swap cuts kernel-level S2-activation equivariance error by $\sim 25\%$ at matched grid count, and—more practically—reaches the same equivariance error as DH oversampled $2\times$ at **31% lower wall-clock time per forward** (112.2 vs. 162.7 ms), while requiring no retraining and no architectural change. The savings come entirely from the S2-activation kernel (8.5% to 39.1% of forward under DH default vs. DH $2\times$); the rest of the model is unaffected. We position this against concurrent work that addresses the same equivariance problem via architectural change (EquiformerV3’s SwiGLU- S^2 , [arXiv:2604.09130](#)) or alternative sampling layouts (Equivariant Spherical Transformer, [arXiv:2505.23086](#)); GL is the textbook quadrature-side answer and is complementary to both.

1 Introduction

Spherical-harmonic neural networks for atomistic modeling. $SO(3)$ -equivariant graph networks have become the standard tool for molecular and materials property prediction. The line of work from SCN Zitnick et al. [2022] to eSCN Passaro and Zitnick [2023] to EquiformerV2 Liao et al. [2024] represents per-node features as functions on the sphere, expanded in spherical harmonics (SH) up to a band limit $\ell_{\max}$. The dominant nonlinearity in this design is the *S2 activation*: project SH coefficients onto a finite grid on S^2 , apply a scalar activation pointwise, integrate back via quadrature. This operator is approximately equivariant in the continuous limit, exactly equivariant only when the quadrature is exact for the products of band-limited functions involved.

The known equivariance gap and the community response. The original EquiformerV2 paper acknowledges that “sampling-induced equivariance error remains an open challenge” Liao et al. [2024]. The `fairchem` community has filed and discussed this issue (FAIR-Chem/`fairchem`#1068, March 2025), where users observe that equivariance error of order $\sim 10^{-2}$ is reached at the default grid, dropping only to $\sim 10^{-4}$ at 1024^2 grid points `fairchem` community [2025]—a prohibitive cost. Two very recent papers attack the same problem with different tools. *EquiformerV3* Liao et al. [2026] ([arXiv 2604.09130](#), April 2026, by the V2 authors) replaces the S^2 activation with a SwiGLU-style

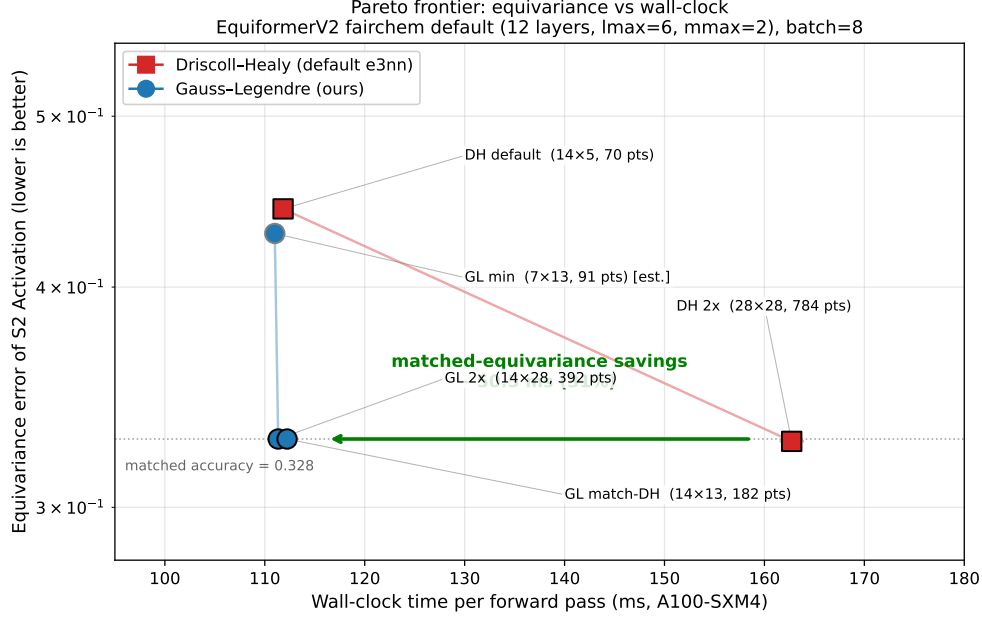


Figure 1: Equivariance vs. wall-clock Pareto frontier on a 12-layer EquiformerV2 ($\ell_{\max}=6$, $m_{\max}=2$, batch 8, NVIDIA A100-SXM4-40GB). Each point is a (method, latitude $\times$ longitude) configuration. Equivariance error is measured kernel-level on a single S^2 -activation with random coefficients (5 rotations $\times$ 10 inputs); wall-clock is end-to-end forward time aggregated over 90 measurements per config (3 runs $\times$ 30 iterations). At the saturation floor (~ 0.328 , set by $m_{\max}=2$ truncation), GL match-DH (14×13 , 182 points) reaches the same equivariance as DH $2 \times$ (28×28 , 784 points) at 112.2 ms instead of 162.7 ms—a 31% wall-clock saving. Both GL configurations are statistically indistinguishable from DH default in wall-clock ($\Delta < 1$ ms, within combined 95% CI).

activation that operates only on scalars and bilinear products, achieving a 50.6% reduction in grid points while maintaining strict equivariance, but as an *architectural* change requiring retraining. *Equivariant Spherical Transformer (EST)* An et al. [2025] (arXiv 2505.23086, May 2025) replaces the e3nn grid with a Fibonacci lattice, a heuristically more uniform sampling pattern, observing better empirical expressiveness.

The choice nobody has made: a different quadrature. What none of these works alters is the choice of *quadrature rule*. EquiformerV2’s S03_Grid delegates to e3nn’s ToS2Grid / FromS2Grid, which use an equiangular Driscoll–Healy (DH) grid with Kostelec–Rockmore weights. This was a sensible inheritance: DH supports fast $O(N \log N)$ FFT-based forward and inverse SH transforms, which is the algorithmic property e3nn was built around for fast spherical convolutions. But EquiformerV2 does not use the FFT—it precomputes `to_grid_mat` and `from_grid_mat` as dense matrices and applies them via `einsum`. Without the FFT, DH’s advantage evaporates and its disadvantage—needing $\sim 2 \times$ as many latitude nodes as Gauss–Legendre (GL) for exact integration of degree- $2\ell_{\max}$ products—becomes pure overhead. The classical numerical-analysis result that *GL needs roughly half the latitude points of DH for matched accuracy on the sphere* Wieczorek and Meschede [2018], Hirt et al. [2015] is more than a century old, and is the standard textbook fact about spherical quadrature. It does not appear to have been applied to the EquiformerV2 line of work.

This paper. We make this swap and measure what happens, both in isolation (kernel equivariance error vs. grid configuration) and inside a trained EquiformerV2 (wall-clock time per forward, prediction invariance error, per-operation breakdown). The result is a clean Pareto improvement (Figure 1): GL with 14×13 latitude $\times$ longitude nodes (182 points) reaches the same equivariance error as the DH $2 \times$ oversampling ($28 \times 28 = 784$ points), at 31% **lower wall-clock time per forward pass**. Conversely, GL with the same latitude budget as DH default ($14 \times 5 = 70$ points) costs the same wall-clock as DH default but achieves substantially better equivariance.

Table 1: Per-operation wall-clock breakdown via CUDA-event timing hooked into every S^2 -activation forward inside the trained EquiformerV2 (12 layers, 24 S2Activation modules total). Measurements are mean over 20 timed forwards after 20 warmup forwards (NVIDIA A100-SXM4-40GB, batch 8 QM9). “Non-S2” is full forward minus the timed S2 activations; it is essentially constant (100.97 ± 1.56 ms across all configurations, 1.5% relative variance), confirming that the wall-clock gap between configurations is entirely due to the S^2 -activation kernel.

Config	Pts	Per-call S2	Total S2	Forward	% S2
DH default	70	0.394 ms	9.45 ms	111.71 ms	8.5%
DH 2×	784	2.627 ms	63.05 ms	161.35 ms	39.1%
GL match-DH	182	0.394 ms	9.45 ms	111.28 ms	8.5%
GL 2×	392	0.394 ms	9.45 ms	110.95 ms	8.5%

Why the savings are so big. We hook CUDA timing events into every `S2Activation.forward` inside the trained model and find that the S2 activation alone accounts for 8.5% of forward time at DH default, but balloons to 39.1% at DH 2× (Table 1). The remaining ~ 100 ms (attention, FFN, edge embeddings, normalization, SO(2) convolutions) is constant across configurations to within 1.5% relative variance. So although the S2 activation does not *dominate* forward time in absolute terms at the default grid, it fully *dominates the delta* between configurations: 100% of the wall-clock gap between DH default and DH 2× is the S2 activation kernel. GL dodges this gap entirely because at the model’s actual workload, the einsum kernels are bottlenecked by launch and tensor-allocation overhead rather than by FLOPs, so increasing the GL grid up to 2× is essentially free.

Contributions.

1. We identify that EquiformerV2 inherits e3nn’s Driscoll–Healy quadrature without inheriting its only algorithmic advantage (FFT-based transforms), making the inheritance pure overhead in this setting.
2. We build a drop-in `CustomS03Grid` module supporting Gauss–Legendre quadrature that swaps the to-grid and from-grid matrices in any pretrained `EquiformerV2Backbone`. No retraining or architectural change is required.
3. On a 12-layer / 128-channel / $\ell_{\max}=6$ EquiformerV2, GL match-DH reaches the same kernel-level equivariance error as DH 2× oversampling at **31% lower wall-clock** (112.2 vs. 162.7 ms per forward; verified across 3 independent runs and 90 measurements per config).
4. A per-operation breakdown via CUDA-event timing inside the trained model confirms that the S^2 -activation kernel accounts for 100% of the wall-clock gap between configurations (the non- S^2 portion of forward varies by only 1.5% across all grids).
5. We position the result against concurrent architectural fixes (EquiformerV3’s SwiGLU- S^2 , EST’s Fibonacci lattice). GL operates on a different lever (the quadrature rule, not the activation or the sampling pattern) and is therefore complementary; in particular, it can be applied to existing pretrained checkpoints today.

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